

PROJECT ABSTRACT

Project Name: **REDESIGNING WEBSITE**

Name of Project Team Members:

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Project Guide: ANIK PATRA

Technology Used: MERN STACK

Platform used: VS CODE

Project Abstract:-

This project presents a strategic initiative to re-engineer and modernize the institution's existing college website by architecting and implementing a full-stack solution utilizing the MERN (MongoDB, Express.js, React.js, Node.js) technology stack. The primary impetus for this overhaul is to address the functional limitations and outdated user interface of the current platform, which impede effective student engagement and administrative efficiency. The project's core objective is to deliver a robust, scalable, and user-centric digital experience for all stakeholders.

The redevelopment encompasses a complete UI/UX transformation, focusing on a responsive and intuitive

design to ensure seamless accessibility across desktops, tablets, and mobile devices. A pivotal component of this project is the introduction of a secure, multi-functional "Student Corner." This dedicated portal will feature an integrated payment gateway for streamlined online fee submissions and a comprehensive system for managing extracurricular activities, from event discovery and registration to participation tracking.

To further enhance user support and communication, an AI-driven chatbot will be integrated to provide 24/7, instantaneous responses to frequently asked questions and user queries, thereby reducing the operational load on administrative staff. The backend, powered by Node.js and Express.js, will expose a set of RESTful APIs to handle all business logic, while MongoDB will serve as the flexible, NoSQL database for efficiently managing complex data structures related to student profiles, financial transactions, and activities. The React.js front-end will consume these APIs to create a dynamic, single-page application (SPA) that ensures a fast and fluid user experience. Ultimately, this technological upgrade aims to not only enhance the college's digital presence but also to streamline critical administrative processes and foster a more connected and interactive campus community.